

HOMEWORK 8 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

E. A. ATINDAMA, PHD MATHEMATICS

Due at 10:00 EST in class

Q1 Solve the following using the superposition method:

(a) $y'' + y = 5 \cos(2x)$

Solution Homogeneous solution:

$$y_h = C_1 \cos x + C_2 \sin x.$$

Particular solution, try

$$y_p = A \cos(2x) + B \sin(2x).$$

Substitute:

$$y_p'' + y_p = (-3A) \cos(2x) + (-3B) \sin(2x) = 5 \cos(2x)$$

so $A = -\frac{5}{3}$, $B = 0$. Hence

$$y = C_1 \cos x + C_2 \sin x - \frac{5}{3} \cos(2x)$$

(b) $y'' - y = xe^x$

Solution

$$y_h = C_1 e^x + C_2 e^{-x}.$$

RHS is (linear polynomial) $\cdot e^x$. Since e^x is a solution of the homogeneous equation (root $r = 1$ of multiplicity 1), multiply the usual trial by x . Trial:

$$y_p = x(Ax + B)e^x = (Ax^2 + Bx)e^x.$$

Compute (as in text) to find $A = \frac{1}{4}$, $B = -\frac{1}{4}$. Thus

$$y = C_1 e^x + C_2 e^{-x} + \frac{x^2 - x}{4} e^x.$$

(c) $y'' + 4y' + 4y = e^{-2x} + x$

Solution:

Homogeneous solution (double root ($r=-2$)):

$$y_h = (C_1 + C_2x)e^{-2x}.$$

Treat RHS by superposition.

For e^{-2x} : resonance with root $r = -2$ of multiplicity 2 so multiply by x^2 .
Trial

$$y_{p1} = Ax^2e^{-2x}.$$

Applying $(D + 2)^2$ gives $2Ae^{-2x}$, so $A = \frac{1}{2}$; hence $y_{p1} = \frac{1}{2}x^2e^{-2x}$.

For polynomial x , try

$$y_{p2} = ax + b.$$

Substituting yields $a = \frac{1}{4}, b = -\frac{1}{4}$

Combine

$$y = (C_1 + C_2x)e^{-2x} + \frac{1}{2}x^2e^{-2x} + \frac{1}{4}x - \frac{1}{4}.$$

Q2 For each of the following, determine the homogeneous solution and the **form** (do not solve) of a particular non-homogeneous solution:

- (i) $y'' + 4y = 2x^2$
- (ii) $y'' + 4y = 10 + \sin(x)$
- (iii) $y'' + 4y = \sin(2x)$
- (iv) $y'' + 4y = e^x \sin(2x)$

Solution

(i) Particular *form*: polynomial of degree 2,

$$y_p = Ax^2 + Bx + C.$$

(ii) y_p is sum of a constant and a sinusoid:

$$y_p = A + (B \cos x + C \sin x).$$

(iii) Particular: sinusoid,

$$y_p = A \cos(2x) + B \sin(2x).$$

(iv) Particular: exponential times sinusoid,

$$y_p = e^x (A \cos(2x) + B \sin(2x)).$$

Q3 Find the general solution of $y'' + 4y = x \cos x$ using the variation of parameters approach.

Solution

Find a inhomogeneous solution of

$$y'' + 4y = x \cos x$$

by variation of parameters.

The homogeneous equation $y'' + 4y = 0$ has fundamental solutions

$$y_1 = \cos 2x, \quad y_2 = \sin 2x,$$

with Wronskian

$$W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} = 2(\cos^2 2x + \sin^2 2x) = 2.$$

Variation of parameters gives

$$u_1'(x) = -\frac{f(x)y_2(x)}{W} = -\frac{x \cos x \sin 2x}{2}, \quad u_2'(x) = \frac{f(x)y_1(x)}{W} = \frac{x \cos x \cos 2x}{2},$$

so

$$u_1(x) = -\frac{1}{2} \int x \cos x \sin 2x \, dx, \quad u_2(x) = \frac{1}{2} \int x \cos x \cos 2x \, dx.$$

Use product-to-sum identities:

$$\cos x \cos 2x = \frac{1}{2}(\cos 3x + \cos x), \quad \cos x \sin 2x = \frac{1}{2}(\sin 3x + \sin x).$$

Thus

$$u_2'(x) = \frac{x}{4}(\cos 3x + \cos x), \quad u_1'(x) = -\frac{x}{4}(\sin 3x + \sin x).$$

Integrate termwise (integration by parts for $\int x \cos kx \, dx$ and $\int x \sin kx \, dx$):

$$\int x \cos kx \, dx = \frac{x \sin kx}{k} + \frac{\cos kx}{k^2}, \quad \int x \sin kx \, dx = -\frac{x \cos kx}{k} + \frac{\sin kx}{k^2}.$$

Hence (dropping arbitrary constants)

$$u_2(x) = \frac{1}{4} \left[\frac{1}{3}x \sin 3x + \frac{1}{9} \cos 3x + x \sin x + \cos x \right],$$

$$u_1(x) = \frac{1}{4} \left[\frac{1}{3}x \cos 3x - \frac{1}{9} \sin 3x + x \cos x - \sin x \right].$$

Therefore a particular solution is

$$y_p(x) = u_1(x) \cos 2x + u_2(x) \sin 2x,$$

i.e.

$$y_p(x) = \frac{1}{4} \left(\left[\frac{1}{3}x \cos 3x - \frac{1}{9} \sin 3x + x \cos x - \sin x \right] \cos 2x + \left[\frac{1}{3}x \sin 3x + \frac{1}{9} \cos 3x + x \sin x + \cos x \right] \sin 2x \right)$$